Frequency-Spatial Dual-Level Selective Alignment for Domain-Adaptive Object Detection in Remote Sensing

Tingting Qiao, He Chen, Jue Wang, *Member, IEEE*, Aijia Shen, Wenchao Liu, Long Pang, and Yizhuang Xie

***Abstract*—Remote sensing object detection suffers from severe performance degradation under cross-domain transfer, where domain gaps arise from differences in spectral response, spatial resolution, and viewing geometry. Most existing unsupervised domain-adaptive object detection (DAOD) methods pursue cross-domain invariance through feature distribution alignment but encounter two limitations specific to remote sensing: cross-domain appearance variation, where differences in imaging conditions produce divergent visual appearances, and foreground-background imbalance, where targets are sparsely distributed across vast backgrounds and alignment is dominated by background statistics. To address these two limitations, frequency-spatial dual-level selective alignment (FS^{2}A), a framework with two complementary modules, is proposed. At the frequency level, scale-class-conditioned frequency modulation (SCFM) decomposes multiscale features via FFT and selectively modulates the low-frequency amplitude conditioned on scale level and categorical composition, restricting adversarial alignment to domain-variant spectral components while preserving the domain-invariant phase spectrum. At the spatial level, kernel relational distillation (KRD) distills pairwise relational structure from a frozen satellite-pretrained vision foundation model (VFM) in polynomial kernel space, where polynomial kernel functions preferentially concentrate alignment on foreground feature pairs over weakly correlated background pairs. Both modules are decoupled from the detection forward pass, introducing no additional inference cost. Experiments on two cross-domain remote sensing benchmarks demonstrate that FS^{2}A achieves 67.6% mAP$_{50}$ on xView→DOTA, surpassing the state-of-the-art by 2.7%, and competitive results on satellite-to-UAV benchmarks. The code will be available at https://github.com/sparklejojo/FSSA-DAOD**

***Index Terms*—Domain adaptation, object detection, remote sensing imagery.**

I. Introduction

REMOTE sensing object detection plays a crucial role in urban planning, disaster response, maritime surveillance, and environmental monitoring [1], [2]. Recent advances in deep learning have substantially improved remote sensing image analysis across challenging scenarios [3], [4], [5]. However, the remote sensing field spans numerous satellite and UAV platforms with diverse spectral responses, spatial resolutions, and viewing geometries. These variations create severe domain gaps, causing detectors trained on one platform to suffer significant performance degradation when deployed on another. Meanwhile, annotating remote sensing imagery requires specialized domain expertise, and the wide diversity of platforms makes cross-platform labeling prohibitively costly. Unsupervised domain-adaptive object detection (DAOD) has therefore emerged as a practical solution, learning from labeled source-domain data and unlabeled target-domain data to achieve robust detection performance despite distribution shifts.

Most existing DAOD methods pursue cross-domain invariance through feature distribution alignment, typically realized via adversarial training with gradient reversal in CNN-based detectors [6], [7], [8]. Transformer-based detectors have more recently enabled multilevel cross-domain alignment within encoder–decoder architectures [9], [10]. RT-DATR [11] represents the current state-of-the-art, integrating adversarial alignment at the local object level, scene semantic level, and instance level within the RT-DETR framework [12]. Yet, as illustrated in Fig. 1, RT-DATR still produces missed detections under cross-domain transfer in remote sensing, where the source Fig. 1(a) and target Fig. 1(b) domains share the same categories, but the detector Fig. 1(c) exhibits degraded performance on the target domain. This performance degradation stems from two limitations intrinsic to remote sensing that existing alignment methods do not fully resolve: cross-domain appearance variation, where differences in imaging conditions produce divergent visual appearances across domains, and foreground-background imbalance, where the extreme sparsity of targets allows background statistics to dominate global alignment.

For cross-domain appearance variation, frequency-domain analysis provides an effective means for characterizing the underlying discrepancy. The Fourier transform, as its primary analytical tool, decomposes features into amplitude and phase spectra. Cross-domain appearance variations, such as differences in illumination and style, predominantly reside in the low-frequency amplitude [13], [14], while the phase spectrum encodes spatial structure that remains stable across domains. Fig. 1(d) corroborates this frequency-domain characterization.

Received 16 April 2026; revised 21 June 2026; accepted 15 July 2026. Date of publication 23 July 2026; date of current version 3 August 2026. This work was supported by the National Natural Science Foundation of China under Grant 62341130. *(Corresponding author: Yizhuang Xie.)*

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Digital Object Identifier 10.1109/TGRS.2026.3716352

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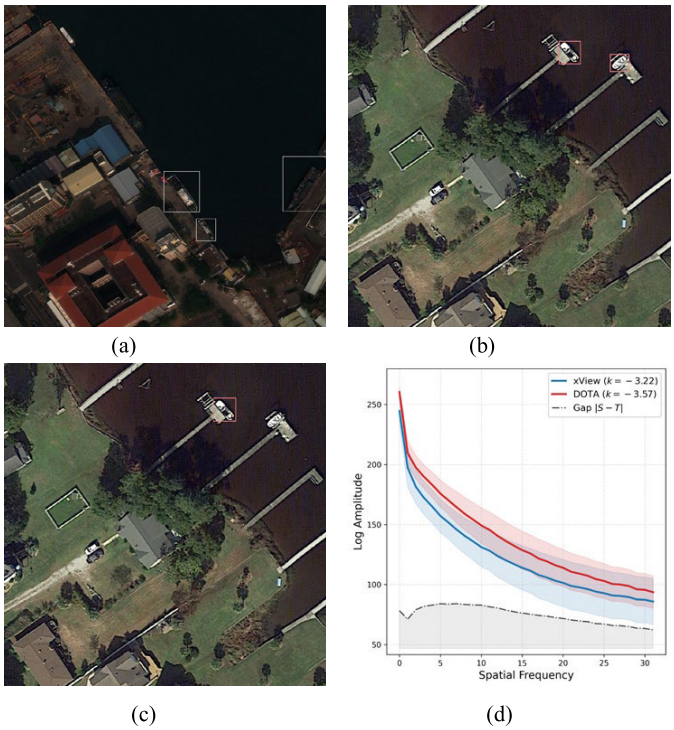


Fig. 1. Cross-sensor domain shift in remote sensing object detection. (a) Source-domain imagery from xView. (b) Target-domain GT from DOTA. (c) Detection results of RT-DATR on the target domain, exhibiting missed detections under domain shift. (d) RAPS of the *Ship* category across both domains. The amplitude gap $|S - T|$ illustrates the spectral discrepancy, and the spectral decay rate k quantifies the frequency distribution.

The radially averaged power spectrum (RAPS) reveals a significant spectral discrepancy between xView and DOTA, concentrated in the low-to-mid frequency range where cross-domain appearance differences are most pronounced. The distinct spectral decay rates (slopes of −3.22 versus −3.57) further quantify the different spectral energy distributions of the two platforms. Existing alignment methods, however, do not differentiate among frequency components and apply equal optimization pressure to amplitude and phase alike. While correcting appearance differences, these methods simultaneously perturb the phase spectrum, unintentionally distorting the spatial structure they should preserve.

To address cross-domain appearance variation, scale-class-conditioned frequency modulation (SCFM) is introduced, which achieves frequency-level selective alignment by decomposing multiscale features via FFT and selectively modulating the low-frequency amplitude conditioned on scale level and categorical composition, restricting adversarial alignment to domain-variant spectral components while protecting the spatial structure encoded in the phase. The scale-class conditioning enables SCFM to apply category-aware and scale-adaptive modulation so that features at different scales and with different categorical distributions receive correction strengths matched to their respective appearance differences rather than a single uniform adjustment.

Beyond cross-domain appearance variation, a distinct challenge arises from foreground-background imbalance. Existing DAOD methods [15], [16] typically rely on adversarial training to align the global feature distributions of the source and target domains. In remote sensing, where foreground targets are sparsely distributed across vast backgrounds, this global alignment is overwhelmingly dominated by background statistics, displacing sparse foreground representations and progressively corrupting their pairwise relational structure during adaptation [17]. Vision foundation models (VFMs) serve as an external structural anchor against this degradation, as their self-supervised representations naturally encode cross-domain semantic similarities [18], [19]. However, directly distilling these VFM features globally remains problematic: the loss is still dominated by abundant background pixels, and mimicking abstract semantic features imposes rigid constraints that compromise precise localization [20], [21].

To address foreground-background imbalance, kernel relational distillation (KRD) is introduced, which enforces spatial-level selective alignment by distilling pairwise relational structure between a frozen satellite-pretrained VFM and the detector in kernel space [22]. Instead of matching individual feature vectors, KRD aligns pairwise relational structure in polynomial kernel space, preferentially concentrating alignment on foreground feature pairs while suppressing background correlations. This kernel-space formulation preserves the geometric capacity of the detection feature space while enabling the flexible transfer of foreground relational structure. In addition, an alignment-state-driven dynamic weighting scheme gradually reduces the constraint strength as alignment converges, preventing interference with detection optimization in the later training stages.

Together, SCFM and KRD are integrated into frequency-spatial dual-level selective alignment (FS^2A), a domain adaptation framework built on RT-DATR. SCFM and KRD are optimized in parallel rather than sequentially: SCFM regularizes an auxiliary adversarial alignment path after frequency modulation to reduce appearance discrepancy, whereas KRD regularizes a separate teacher–student relational path to preserve foreground relational structure. Both paths are decoupled from the primary detection path and affect the shared model parameters only through back-propagation, introducing no additional inference cost. Their effects are complementary because frequency-level modulation reduces domain-variant appearance differences, while spatial-level relational distillation prevents sparse foreground structure from being overwhelmed by background-dominated alignment.

The main contributions of this work are summarized as follows.

1) To address cross-domain appearance variation and foreground-background imbalance in existing alignment methods for remote sensing DAOD, FS^2A, a dual-level selective alignment framework built on RT-DATR with two complementary modules, SCFM and KRD, is proposed. Both modules selectively target domain-variant components at each level while preserving domain-invariant structure and are fully decoupled from the detection forward pass, introducing no additional inference cost.
2) To address cross-domain appearance variation, SCFM is proposed, which achieves frequency-level selective

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alignment by decomposing multiscale features via FFT and selectively modulating the low-frequency amplitude conditioned on scale level and categorical composition. By restricting adversarial alignment to domain-variant spectral components while preserving the phase spectrum that encodes spatial structure, SCFM achieves category-aware and scale-adaptive frequency alignment.

3) To address foreground-background imbalance, KRD is proposed, which enforces spatial-level selective alignment by distilling pairwise relational structure from a frozen satellite-pretrained VFM in polynomial kernel space. The polynomial kernel functions preferentially concentrate alignment on foreground feature pairs over weakly correlated background pairs, achieving foreground-focused relational alignment. An alignment-state-driven dynamic weighting scheme gradually reduces the constraint strength as alignment converges, preventing the relational loss from dominating optimization in later training stages.

Experiments on two cross-domain remote sensing benchmarks, covering cross-satellite (xView → DOTA) and satellite-to-UAV (UCAS-AOD ↔ CARPK) transfers, demonstrate that FS^{2}A achieves state-of-the-art performance on the cross-satellite benchmark and competitive results on satellite-to-UAV benchmarks, validating the effectiveness of its dual-level selective alignment design.

The remainder of this article is organized as follows. Section II reviews related work on DAOD and VFMs. Section III presents the FS^{2}A framework in detail. Section IV reports experimental results and ablation studies. Section V concludes this article.

II. Related Work

A. Domain-Adaptive Object Detection

The foundational paradigm in DAOD employs adversarial feature alignment to bridge cross-domain discrepancies. Early methods achieve this through gradient reversal layers, enforcing image-level and instance-level invariance [6], [23]. To mitigate counterproductive uniform alignment across globally dissimilar image pairs, subsequent approaches apply hierarchical alignment across different semantic granularities [7]. Furthermore, methodological refinements decouple tasks during cross-domain alignment [24], utilize graph-based prototype matching [25], and impose categorical regularization constraints [16]. As a complementary paradigm to adversarial training, self-training methods leverage pseudolabel refinement via mean-teacher architectures [26], subsequently adapted for dense object detection through unbiased pseudolabeling [27] and soft-weighting strategies [28]; recent work further reinforces these methods using contrastive learning objectives [29].

The adoption of Transformer-based architectures shifts the alignment paradigm toward attention-guided mechanisms within encoder–decoder frameworks. These methods align features token by token using domain queries [9] or impose spatial, channel, and instance-level constraints via adversarial tokens [10]. Other approaches mitigate domain shift explicitly at the backbone [30], fuse complementary CNN and transformer features [31], or rely on masked feature reconstruction to capture target-domain semantics under limited supervision [32], [33]. Recent real-time architectures integrate multilevel feature alignment directly into the RT-DETR pipeline [11], [12].

B. DAOD in Remote Sensing

The application of DAOD to remote sensing imagery encounters compounded challenges absent from standard natural image benchmarks: sparsely distributed foreground targets against visually cluttered backgrounds and pronounced cross-domain shift arising from heterogeneous sensor spectral responses and varying viewing geometries.

To address foreground-background confounding and target sparsity, methodologies largely focus on selective spatial constraints. Prominent strategies decouple foreground from background using adversarial constraints combined with prototype-based alignment [8], utilize generative adversarial transfer coupled with hard example mining to filter and refine pseudolabels [34], or impose prospective consistency constraints to manage dense and multiscale geospatial objects during self-training [35]. Other spatial-domain interventions employ generative image translation to explicitly close the visual gap before feature extraction [36].

To mitigate profound spectral shifts, a parallel trajectory of work has initiated a transition from purely spatial techniques to frequency-domain operations. Building on established techniques such as FDA [13], which performs simple image-level low-frequency amplitude swapping to align styles, recent methods adapt frequency transformations for object detection. For example, FIE-Net [14] preprocesses images using FFT-based style transfer before applying multigranularity spatial alignment. Within transformer architectures, remote sensing teacher (RST) [37] projects multiscale features into the frequency domain to extract domain-invariant components in a DETR framework.

Despite this progress, these frequency-domain methods apply spectral operations uniformly. Methods like FDA and FIE-Net restrict their frequency interventions to global, image-level preprocessing, while feature-level approaches like RST align projected frequency components indiscriminately. Remaining entirely agnostic to the distinct spectral characteristics of different object categories and feature pyramid (FPN) scales, such undifferentiated frequency modulation fails to exploit the unique spectral signatures associated with each category and scale level. This leaves category-aware, scale-adaptive frequency alignment as a critical unresolved challenge in remote sensing DAOD.

C. VFMs for Detection and Adaptation

VFMs pretrained via self-supervised objectives on large-scale corpora have established a new standard for cross-domain generalization in feature representation [38]. The progression of the DINO family illustrates this methodological shift: initial architectures pioneered self-distillation without labels to yield semantically structured visual features [18],

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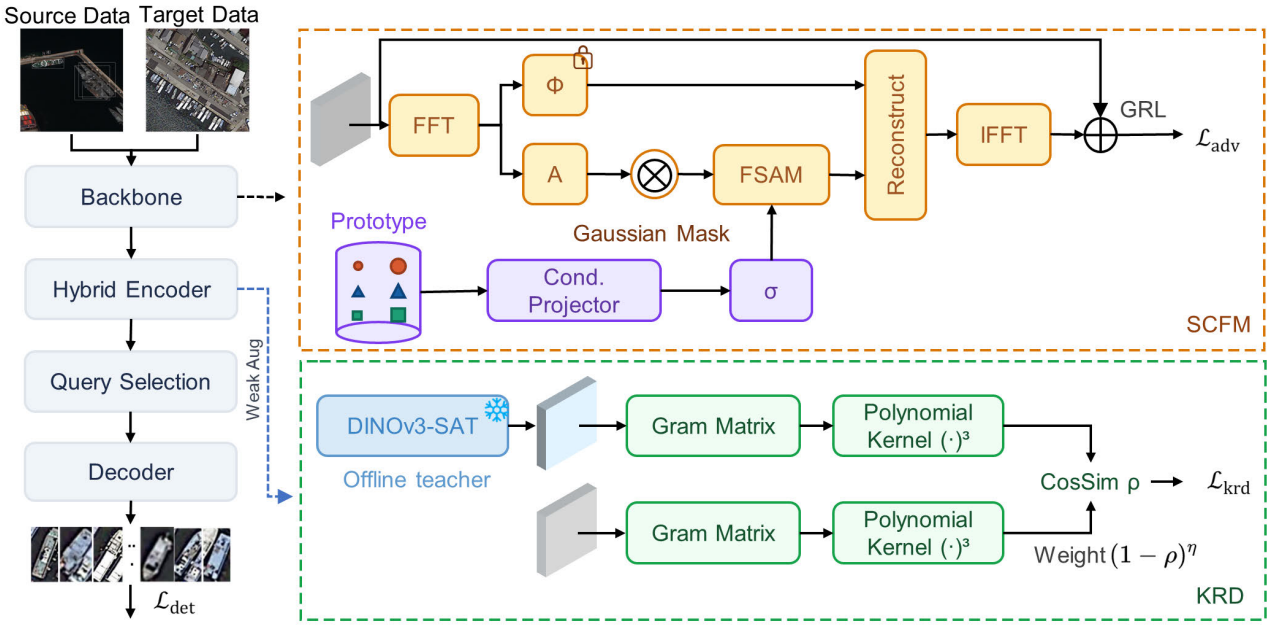


Fig. 2. Overview of FS^{2}A, a dual-level framework for DAOD. Two parallel regularization paths are introduced: SCFM modulates low-frequency amplitude under scale-class conditioning for adversarial alignment; KRD aligns pairwise relational structure with a frozen DINOv3-SAT teacher in polynomial kernel space via a dual-stream forward pass.

which subsequently scaled to curated datasets to produce robust representations across diverse visual domains [39]. The recent extension of this paradigm into domain-specific pretraining introduces specialized variants like DINOv3-SAT [40]. Trained exclusively on the SAT-493 M corpus, this domain-specific VFM generates tailored geospatial representations that consistently outperform general-purpose models on remote sensing benchmarks.

The strong transferability of these representations motivates leveraging VFMs as external guidance for cross-domain alignment. Existing strategies distill vision-language knowledge via domain-mixed contrastive constraints to improve adaptation under limited target-domain labels [41]. However, directly integrating VFM representations into remote sensing detection pipelines exposes two critical methodological challenges. First, dense background regions overwhelm global feature similarity computations, causing the relational structure among sparse foreground targets to be underrepresented in the alignment signal. Second, VFM representations inherently encode abstract, semantic-level generalizations that impose rigid constraints. Such abstract representations fundamentally conflict with the precise geometric localization gradients required by the detection head, impeding late-stage spatial optimization. These conflicting requirements dictate the need for alignment mechanisms capable of isolating relational knowledge from dense backgrounds without perturbing the geometric capacity of the detection pipeline.

III. FS^{2}A Framework

A. Overview

1) Problem Formulation: The standard unsupervised DAOD setting is considered. A labeled source domain $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ and an unlabeled target domain $\mathcal{D}_t = \{x_j^t\}_{j=1}^{N_t}$ are provided, where x_i^s and x_j^t are input images, and y_i^s contains ground-truth bounding boxes and category labels. Both domains share the same category set but exhibit significant appearance discrepancies due to differences in sensor spectral response, spatial resolution, and viewing geometry. The goal is to train a detector on $\mathcal{D}_s$ that generalizes well to $\mathcal{D}_t$.

2) Framework Overview: As analyzed in Section I, applying existing DAOD methods to remote sensing exposes two limitations: cross-domain appearance variation, where differences in imaging conditions produce divergent visual appearances across domains, and foreground-background imbalance, where the extreme sparsity of targets allows background statistics to dominate alignment. FS^{2}A addresses both through two complementary and parallel modules: SCFM for frequency-level selective alignment and KRD for spatial-level selective alignment. The overall architecture is illustrated in Fig. 2.

The base detector follows RT-DETR, which consists of a ResNet backbone, a hybrid encoder, and a DETR-style decoder. The hybrid encoder first projects multiscale backbone features to a uniform channel dimension, then applies a Transformer encoder to the coarsest-scale features for global context modeling, and finally fuses the results through a bidirectional aggregation path that combines top-down semantic propagation and bottom-up spatial refinement, producing three multiresolution output feature maps.

SCFM operates on the channel-projected multiscale features at the input of the hybrid encoder. For each scale, a 2-D Fourier transform is applied to decompose the feature map into amplitude and phase, which are then used to selectively modulate the low-frequency amplitude under a conditioning vector that encodes the scale level and categorical composition of the current feature, while the phase spectrum remains intact.

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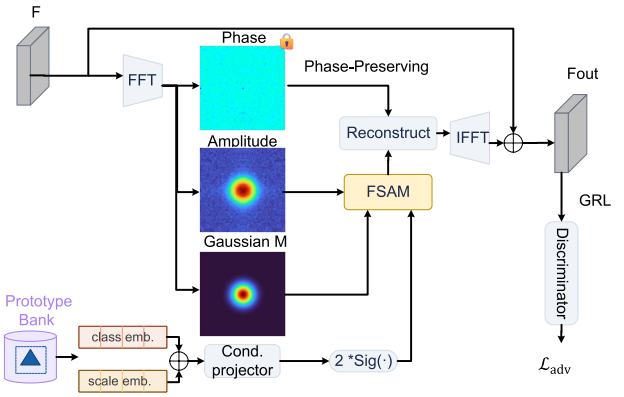


Fig. 3. Architecture of SCFM. Features are decomposed via FFT; a scale-class-conditioning vector drives per-channel modulation factors α through an MLP. A Gaussian mask restricts modulation to low frequencies while the phase spectrum is kept intact. The modulated feature is fed into a domain discriminator through a gradient reversal layer.

KRD operates on the middle-resolution output of the hybrid encoder. It distills pairwise relational structure in polynomial kernel space [42] from a frozen satellite-pretrained VFM via a dual-stream forward pass, preserving the geometric capacity of the detection feature space while enabling the flexible transfer of foreground relational structure; the superlinear response of the polynomial kernel preferentially upweights high-similarity foreground token pairs over background pairs.

The two modules do not alter the feature streams used by each other or by the detection head; their effect on the shared model parameters is exerted exclusively through gradient back-propagation. Sections III-B–III-D detail SCFM, KRD, and the overall training objective, respectively.

B. Scale-Class-Conditioned Frequency Modulation

To address cross-domain appearance variation, SCFM achieves frequency-level selective alignment by preserving the phase spectrum while selectively modulating the low-frequency amplitude under scale-class conditioning, as illustrated in Fig. 3. Existing alignment methods treat amplitude and phase as a single undifferentiated signal, applying equal optimization pressure to both simultaneously. Since cross-domain appearance variations are concentrated in the low-frequency amplitude while the phase encodes domain-invariant spatial structure [13], this undifferentiated pressure acts on the phase spectrum, a component that should remain intact. This mismatch is particularly consequential in remote sensing, where cross-domain shifts produce large spectral amplitude discrepancies. SCFM exploits the Fourier decomposition (amplitude $\leftrightarrow$ style; phase $\leftrightarrow$ spatial structure) to decouple the two: a smooth Gaussian mask restricts modulation to the low-frequency amplitude band, while a scale-class-conditioning vector steers per-channel scaling according to the scale level and categorical composition, making the alignment category- and scale-aware. The phase spectrum remains intact throughout, so spatial structure is preserved by construction.

1) Scale-Class-Conditioning Vector: Spectral discrepancy in remote sensing is heterogeneous across feature scales and object categories: features at different scale levels exhibit distinct frequency profiles due to their varying receptive fields and degrees of semantic abstraction, and different object categories carry different dominant spectral signatures. Applying a single shared modulation factor would therefore impose disproportionate correction across the feature hierarchy, overcorrecting some scale-category combinations while undercorrecting others. To avoid blind class-agnostic alignment, a dynamic conditioning vector $\mathbf{v} \in \mathbb{R}^{d_s+K}$ is constructed, where d_s is the scale embedding dimension and K is the number of categories, encoding both the scale level and the categorical composition of each feature map.

The scale component is a learnable embedding indexed by the scale level. For the class component, a multiscale prototype bank is maintained $\mathbf{P} \in \mathbb{R}^{L\times K\times C}$, where L, K, and C denote the number of scale levels, categories, and channels, respectively. $\mathbf{P}$ is initialized as an all-zero buffer and populated online using source-domain samples. To avoid cold-start bias, when the prototype bank is still uninitialized, the first available source-domain box-pooled features are directly assigned to the corresponding prototypes. Once the bank has been initialized, prototypes are updated by exponential moving average with momentum $\mu = 0.99$ using only source-domain ground-truth boxes, avoiding noise from target-domain pseudolabels.

For both the source and target domains, the class embedding is computed identically. At each spatial position, the feature vector is compared against the current-level prototypes via cosine similarity, producing a dense similarity map $\mathbf{S} \in \mathbb{R}^{HW\times K}$. Negative similarities are suppressed by ReLU, and the class embedding is obtained by taking the spatial maximum followed by L2 normalization

$$\mathbf{e}_{\text{class}} = \text{Normalize}\left(\left[\max_{1\le i\le HW} \text{ReLU}\left(\mathbf{S}_{i,k}\right)\right]_{k=1}^{K}\right) \in \mathbb{R}^{K}. \quad (1)$$

Under the shared label space assumption of unsupervised domain adaptation, source-domain prototypes serve as reliable categorical priors for the conditioning vector; the spatial max-pooling aggregation ensures responsiveness when foreground targets are sparsely distributed, as any high-confidence spatial match activates the corresponding class dimension.

The final conditioning vector concatenates the scale and class embeddings

$$\mathbf{v} = \left[\mathbf{e}_{\text{scale}}; \mathbf{e}_{\text{class}}\right] \in \mathbb{R}^{d_s+K}. \quad (2)$$

2) Frequency Selective Amplitude Modulation: Given an input feature map $\mathbf{F} \in \mathbb{R}^{B\times C\times H\times W}$ and the conditioning vector $\mathbf{v}$, the modulation layer first applies a 2-D real FFT to obtain the complex spectrum

$$\hat{\mathbf{F}} = \text{rfft2}\left(\mathbf{F}\right), \quad \mathbf{A} = |\hat{\mathbf{F}}|, \quad \Phi = \angle\hat{\mathbf{F}} \quad (3)$$

where $\mathbf{A}$ is the amplitude and Φ is the phase.

The conditioning vector is then mapped to a per-channel modulation factor through a two-layer MLP

$$\alpha = 2\cdot\sigma\left(\text{MLP}\left(\mathbf{v}\right)\right) \in \mathbb{R}^{B\times C\times 1\times 1} \quad (4)$$

where $\sigma(\cdot)$ denotes the sigmoid function. The modulation factor α has a range of $[0, 2]$, centered at 1.0 (identity transform). When $\alpha_c > 1$, the spectral energy in channel c is amplified; when $\alpha_c < 1$, it is suppressed. This identity-centered

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modulation ensures that SCFM defaults to an identity mapping when the conditioning signal is uninformative, for example, during early training before prototypes stabilize.

To restrict modulation to the low-frequency band, a smooth Gaussian mask $\mathcal{M}$ is constructed over the frequency plane. For the 2-D real FFT output, the DC component is located at $(0, 0)$, and the vertical frequency index is periodic. We therefore compute the minimum squared distance to the nearest low-frequency origin under periodic vertical indexing

$$d^2(u, v) = \min\left(u^2 + v^2,\ \left(H_f - u\right)^2 + v^2\right) \tag{5}$$

and define the smooth frequency-domain mask as

$$\mathcal{M}(u, v) = \exp\left(-\frac{d^2(u, v)}{2\left(r \cdot W_f\right)^2}\right) \tag{6}$$

where r is the low-frequency ratio, and W_f is the frequency-domain width. The mask approaches 1 at low frequencies and decays smoothly to 0 at high frequencies, avoiding the ringing artifacts that a hard binary cutoff would introduce.

The modulated amplitude is computed as a frequency-selective blending between the original and scaled amplitudes

$$\tilde{\mathbf{A}} = \mathbf{A} \odot (1 - \mathcal{M}) + (\mathbf{A} \odot \alpha) \odot \mathcal{M}. \tag{7}$$

This formulation ensures that low-frequency amplitudes are modulated by the conditioning-derived factor α, while high-frequency amplitudes remain unmodified, and the transition is smooth. The phase spectrum Φ is preserved throughout the entire process without any modification.

The modulated feature is then reconstructed via inverse FFT from the modified amplitude and the original phase

$$\tilde{\mathbf{F}} = \text{irfft2}\left(\tilde{\mathbf{A}} \cdot e^{j\Phi};\ s = (H, W),\ \text{norm} = \text{ortho}\right). \tag{8}$$

The final output employs a residual connection with a learnable scalar β initialized to zero

$$\mathbf{F}_{\text{out}} = \mathbf{F} + \beta\tilde{\mathbf{F}}. \tag{9}$$

With zero initialization of β, SCFM starts as an identity mapping by design, which stabilizes early optimization.

The modulated features $\mathbf{F}_{\text{out}}$ are then fed into a domain discriminator through a gradient reversal layer for adversarial training. Through gradient reversal, the backbone is encouraged to produce features whose low-frequency amplitude is domain-indistinguishable, while the phase-encoded spatial structure remains protected from adversarial perturbation. The resulting adversarial loss $\mathcal{L}_{\text{adv}}^{B}$ is detailed in Section III-D.

C. Kernel Relational Distillation

To address foreground-background imbalance, KRD enforces spatial-level selective alignment by distilling pairwise relational structure [22] from a frozen satellite-pretrained VFM in polynomial kernel space, rather than matching individual feature vectors, as shown in Fig. 4. This kernel-space formulation preserves the geometric capacity of the detection feature space while enabling the flexible transfer of foreground relational structure. Global

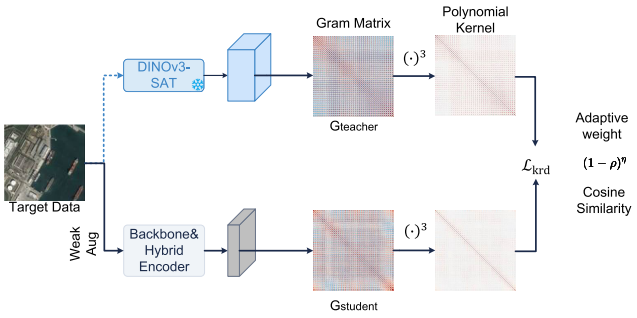


Fig. 4. Architecture of KRD. A weakly augmented target image is forwarded with frozen BN to extract student features. Tokens are mean-centered and L2-normalized; pairwise cosine similarity matrices are transformed into degree-3 polynomial kernel matrices. The cosine similarity ρ between the student and teacher kernel matrices drives the dynamic weighting $(1 - \rho)^\eta$.

feature-level alignment is overwhelmingly dominated by background statistics, as foreground targets are sparsely distributed across vast backgrounds; directly distilling VFM features globally imposes rigid constraints that compromise precise localization [20], [21]. This selectivity is grounded in the semantic structure of DINOv3-SAT [40], a satellite-domain VFM trained exclusively on the SAT-493 M corpus via self-supervised objectives. Its self-supervised training objective encourages semantically coherent representations, such that foreground tokens tend to exhibit higher mutual similarity than background tokens across imaging conditions. KRD exploits the superlinear response of the polynomial kernel to reweight pairwise token affinities based on this property. The odd degree preserves the sign of pairwise similarity, while the cubic exponent preferentially upweights foreground-foreground pairs over low-similarity background pairs. This concentrates the alignment signal on foreground relational structure. An alignment-state-driven dynamic weighting scheme gradually reduces the constraint strength as alignment converges, preventing interference with detection optimization in later training stages.

1) Kernel Function and Kernel Matrix: Let $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ denote a similarity function that maps any pair of feature vectors $(\mathbf{x}, \mathbf{y})$ to a scalar score. For an arbitrary sample set $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, the corresponding Gram matrix $\mathbf{K} \in \mathbb{R}^{n \times n}$ is defined by its elements $\mathbf{K}_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$. The function k constitutes a valid kernel provided that this Gram matrix remains positive semidefinite for all finite sample collections [42]. This characteristic guarantees that the pairwise relationships between samples are embedded with geometric consistency.

In this work, a homogeneous polynomial kernel with $\gamma = 1$ and $c = 0$ is employed, parameterized solely by degree d

$$k_{\text{poly}}(\mathbf{x}, \mathbf{y}) = \left(\mathbf{x}^\top \mathbf{y}\right)^d. \tag{10}$$

To eliminate the influence of varying feature magnitudes, a normalized formulation is introduced

$$\tilde{k}(\mathbf{x}, \mathbf{y}) = \frac{k(\mathbf{x}, \mathbf{y})}{\sqrt{k(\mathbf{x}, \mathbf{x})\ k(\mathbf{y}, \mathbf{y})}} \tag{11}$$

which inherently enforces $\tilde{k}(\mathbf{x}, \mathbf{x}) = 1$ and stabilizes the relational distillation process.

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2) Feature Preparation: DINOv3-SAT is adopted as the frozen teacher. For each target-domain image, the ViT patch-token features are extracted offline and saved as spatial feature maps $\mathbf{T} \in \mathbb{R}^{C_t \times H_t \times W_t}$, incurring no additional GPU cost during training.

During training, the target image undergoes a dual-stream forward pass. Strong augmentations are processed for the primary detection loss, whereas weak augmentations are processed for relational alignment. The alignment stream applies only a deterministic resize to 640 $\times$ 640 and normalization statistics matched to DINOv3-SAT's pretraining distribution, with no random spatial transformations. Because both the offline teacher features and the weakly augmented student input are derived from the same fixed-size resize of the original image, their spatial token layouts are in exact correspondence, fully preserving the pairwise relational structure. To obtain the student feature, the weakly augmented image is passed through the backbone and hybrid encoder with all batch normalization layers temporarily set to evaluation mode. This prevents the weak-stream data from contaminating the running statistics of the detection pipeline. The student feature is extracted at the first bottom-up aggregation point of the hybrid encoder. Specifically, we use the satellite-pretrained DINOv3 ViT-L/16 checkpoint trained on SAT-493 M; its final hidden-state output is processed by removing the prefix tokens (one CLS token and four register tokens), and the remaining patch tokens are reshaped into $\mathbf{T} \in \mathbb{R}^{1024 \times 40 \times 40}$ under the fixed 640 $\times$ 640 input resolution. These patch tokens carry globally attended semantic information at a fixed spatial resolution of 40 $\times$ 40; the structural counterpart in the hybrid encoder is this aggregation point, where the Transformer encoder has already injected global cross-scale attention analogous to ViT self-attention, while the bottom-up path preserves the finest spatial granularity. Shallower backbone features lack this cross-scale semantic context and cannot form reliable pairwise affinities for distillation; deeper aggregation stages have reduced spatial resolution that mismatches the teacher token layout. The first bottom-up aggregation point is thus the earliest location at which semantic pairwise structure is sufficiently formed to distill from DINOv3-SAT's globally-attended patch-token affinities.

3) Polynomial Kernel Relational Structure: Given the student feature $\mathbf{F}_s \in \mathbb{R}^{B \times C_s \times H \times W}$ and teacher feature $\mathbf{F}_t \in \mathbb{R}^{B \times C_t \times H \times W}$, we first reshape each into $N = HW$ spatial tokens, denoted $\mathbf{Z}_s, \mathbf{Z}_t \in \mathbb{R}^{B \times N \times C}$, respectively. Spatial mean-centering is performed by subtracting the tokenwise spatial mean, removing the global activation bias that differs across domains. Each centered token is then L2-normalized to unit length (denoted $\hat{\mathbf{Z}}_s$ and $\hat{\mathbf{Z}}_t$ for the student and teacher, respectively), making the subsequent inner product equivalent to cosine similarity. This normalization also enables comparison between the student and teacher despite their different channel dimensions.

The pairwise similarity matrix is computed independently for each stream

$$\mathbf{G}_s = \hat{\mathbf{Z}}_s \hat{\mathbf{Z}}_s^\top, \quad \mathbf{G}_t = \hat{\mathbf{Z}}_t \hat{\mathbf{Z}}_t^\top, \quad \mathbf{G}_s, \mathbf{G}_t \in \mathbb{R}^{B \times N \times N} \tag{12}$$

where each entry $(G_s)_{ij}$ captures the cosine similarity between student tokens i and j, and analogously for $\mathbf{G}_t$. A polynomial kernel is applied elementwise to each similarity matrix, yielding the kernel matrices

$$(K_s)_{ij} = \left((G_s)_{ij}\right)^d, \quad (K_t)_{ij} = \left((G_t)_{ij}\right)^d, \quad d = 3 \tag{13}$$

where the exponent is applied elementwise (Hadamard product), and each entry captures the amplified token affinity between positions i and j. We adopt this kernel for two complementary properties suited to foreground reweighting: the odd degree preserves the sign of pairwise similarity, maintaining semantic polarity across positive and negative token correlations; the cubic exponent provides superlinear amplification, representing the minimal odd degree that effectively distinguishes high-similarity foreground pairs from low-similarity background pairs without introducing learned parameters.

This reweighting is grounded in the VFM's semantic representation structure: foreground regions exhibit higher mutual similarity in the DINOv3-SAT feature space, while background regions, which lack consistent semantic structure, produce comparatively low pairwise correlations. Because the cubic function grows superlinearly near unity and collapses near zero, it substantially amplifies high-similarity foreground pairs while suppressing low-similarity background pairs, concentrating the alignment signal on semantically coherent structure without explicit segmentation labels.

Unlike batchwise relational distillation methods that require large batch sizes to construct meaningful Gram matrices, the proposed spatial-tokenwise formulation computes pairwise relations within each image. This makes it compatible with the small batch sizes typical in detection training. $\mathbf{K}_t$ is detached from the computation graph to prevent gradient flow into the frozen teacher features.

4) Alignment-State-Driven Dynamic Weighting: The alignment state ρ_b is measured in the same kernel space as the optimization objective rather than in the raw feature space. A feature-space cosine similarity would capture overall representational proximity, including domain-specific activation patterns unrelated to pairwise relational structure, and could indicate high alignment even when pairwise relational structures remain inconsistent. By measuring ρ_b in kernel space, the weighting factor directly reflects the relational alignment quality that the loss is designed to improve, ensuring that the self-pacing signal is consistent with the learning objective. For each sample b in the minibatch, ρ_b is computed as the cosine similarity between the flattened student and teacher kernel matrices

$$\rho_b = \frac{\operatorname{vec}\left(\mathbf{K}_s^{(b)}\right)^\top \operatorname{vec}\left(\mathbf{K}_t^{(b)}\right)}{\left\|\operatorname{vec}\left(\mathbf{K}_s^{(b)}\right)\right\|_2 \left\|\operatorname{vec}\left(\mathbf{K}_t^{(b)}\right)\right\|_2}. \tag{14}$$

The unweighted KRD loss is then defined by averaging the per-sample alignment gap

$$\mathcal{L}_{\mathrm{KRD}} = \frac{1}{B} \sum_{b=1}^{B} (1 - \rho_b)^\eta, \quad \eta = 1.5. \tag{15}$$

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This formulation creates a smooth optimization trajectory. In early training, when ρ is low, $(1-\rho)^{1.5}$ provides a strong gradient that establishes the coarse relational structure among foreground regions. As training progresses and $\rho \to 1$, the loss gradient diminishes smoothly, freeing the detection head to focus on precise bounding box regression without continued interference from the relational constraint. The exponent $\eta = 1.5$ provides superlinear decay that progressively reduces the constraint strength as alignment converges. This state-driven weighting eliminates the need for manually designed curricula or heuristic scheduling strategies, allowing the model to automatically adjust the strength of relational alignment based on its current alignment quality.

D. Training Objective

The overall optimization objective consists of three groups: the supervised detection loss, the multilevel adversarial domain alignment loss, and the KRD loss.

The detection loss $\mathcal{L}_{det}$ is computed exclusively on source-domain samples following the standard RT-DETR formulation, which includes classification focal loss [43], bounding box L1 loss, GIoU loss [44], and the corresponding denoising training losses.

Following the multilevel adversarial alignment paradigm of the RT-DATR baseline, domain discriminators are employed at three architectural levels, each integrated with a gradient reversal layer to learn domain-invariant features. At the backbone level, the SCFM-modulated features $\mathbf{F}_{\text{out}}^{(i)}$, described in Section III-B are fed into a pixel-level domain discriminator. The backbone adversarial loss combines a global pixelwise loss $\mathcal{L}_{\text{loc}}$ with an instance-weighted loss $\mathcal{L}_{\text{obj}}$ that emphasizes foreground regions. A spatial weight matrix w_i is constructed with the same spatial size as the feature map f_i, where regions within ground-truth boxes in the source domain or high-confidence pseudolabel boxes in the target domain receive elevated weights, and the remaining regions are assigned baseline weights. In the target domain, high-confidence pseudolabel boxes are defined as the top 100 final-layer query-class predictions whose confidence scores exceed 0.5. The backbone-level adversarial loss is

$$\mathcal{L}_{\text{loc}} = \sum_{i=1}^{M} \sum_{h,w} -d \log D_b\left(f_i^{(h,w)}\right) - (1-d)\log\left(1 - D_b\left(f_i^{(h,w)}\right)\right) \tag{16}$$

$$\mathcal{L}_{\text{obj}} = \sum_{i=1}^{M} \sum_{h,w} -w_i^{(h,w)}\, d \log D_b\left(f_i^{(h,w)}\right) - w_i^{(h,w)}\,(1-d)\log\left(1 - D_b\left(f_i^{(h,w)}\right)\right) \tag{17}$$

$$\mathcal{L}_{\text{adv}}^{B} = \mathcal{L}_{\text{loc}} + \mathcal{L}_{\text{obj}}. \tag{18}$$

Here, d denotes the domain label, D_b represents the backbone domain discriminator, f_i denotes the feature map at the ith FPN scale, and (h, w) indicates the spatial position within the feature map. In the proposed framework, the features f_i fed into the discriminator are the SCFM-modulated features $\mathbf{F}_{\text{out}}^{(i)}$ rather than the raw projected features. This is the integration point where SCFM enters the adversarial alignment: by performing frequency-selective modulation before discrimination, the adversarial gradient encourages the backbone to produce features whose low-frequency amplitude is domain-indistinguishable while the phase-encoded spatial structure remains intact.

At the encoder level, a global discriminator aligns the multiscale encoder outputs for scene-level semantic alignment, producing the loss $\mathcal{L}_{\text{adv}}^{E}$. At the decoder level, instance-level feature alignment is performed using domain queries decoupled from object queries, with focal weighting applied to handle the imbalance between easy and hard domain-classification samples, producing the loss $\mathcal{L}_{\text{adv}}^{D}$. Both losses follow the RT-DATR formulation without modification.

In addition, a cross-layer mean-teacher consistency loss $\mathcal{L}_{\text{cons}}$ is applied to the target-domain decoder outputs. The reference prediction is obtained by averaging predictions from all decoder layers, and consistency is enforced via JS divergence on classification logits and L1 loss on box predictions.

The KRD loss $\mathcal{L}_{\text{KRD}}$ is computed on target-domain features as described in (15).

The total training loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + \lambda_1 \mathcal{L}_{\text{adv}}^{B} + \lambda_2 \mathcal{L}_{\text{adv}}^{E} + \lambda_3 \mathcal{L}_{\text{adv}}^{D} + \lambda_4 \mathcal{L}_{\text{cons}} + \lambda_5 \mathcal{L}_{\text{KRD}} \tag{19}$$

where λ_1 through λ_5 denote the weight coefficients for each loss term. $\mathcal{L}_{\text{adv}}^{B}$ is the sole entry point of SCFM into the optimization, while $\mathcal{L}_{\text{KRD}}$ is the sole entry point of KRD.

The two modules are decoupled in the forward pass: SCFM-modulated features do not enter the feature aggregation network or the detection pipeline, and KRD operates on the postaggregation output of the hybrid encoder. Yet they jointly regularize the shared backbone through their respective gradient paths in the backward pass.

IV. Experiments

A. Datasets

Experiments are conducted under two widely adopted domain adaptation scenarios in remote sensing, covering cross-domain shifts of different types. Four publicly available datasets are employed across these scenarios. The domain settings, category configurations, and data scales are detailed in the following subsections.

1) Cross-Satellite Adaptation (xView → DOTA): This scenario evaluates knowledge transfer across different optical satellite platforms, where domain shift arises from differences in sensor spectral response, spatial resolution, imaging geometry, and annotation granularity. xView [45] is used as the source domain and DOTA [46] as the target domain.

a) DOTA: DOTA is one of the largest publicly available aerial remote sensing datasets, comprising 2806 large-scale images collected from multiple platforms, including Google Earth and aerial cameras. It contains 188 282 annotated instances spanning 15 object categories. Three categories that overlap with xView are selected: airplane, ship, and storage tank. Images are cropped into 800 × 800 patches with an overlapping stride of 100 pixels, and training and evaluation follow the official trainval split.

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b) xView: The xView dataset is collected by the WorldView-3 satellite at a 0.3 m ground sample distance, providing high-resolution coverage of diverse global scenes with over one million fine-grained bounding box annotations across 60 object categories. Because xView adopts a highly granular taxonomy relative to DOTA, semantically related subcategories are merged to align the label spaces (e.g., motorboat, sailboat, and fishing vessel are merged into the ship category; fixed-wing aircraft subcategories are merged into airplane). After merging, three common categories are retained for adaptation experiments. Images are cropped to 800×800 patches with a stride of 100 pixels, consistent with DOTA preprocessing.

2) Satellite-to-UAV Adaptation (UCAS-AOD ↔ CARPK): This scenario evaluates cross-platform generalization for vehicle detection, where domain shift is caused by differences in imaging altitude, viewpoint angle, object scale, and scene composition between satellite and UAV imagery. Experiments are conducted in both transfer directions using UCAS-AOD [47] and CARPK [48], focusing exclusively on the car category.

a) UCAS-AOD: UCAS-AOD is a remote sensing object detection dataset collected from Google Earth by the University of Chinese Academy of Sciences. It contains 510 images at a resolution of 1280×659 pixels, with 7114 annotated vehicles exhibiting a relatively uniform distribution of orientations. As no official train/test split is provided, a reproducible partitioning strategy is adopted: images whose filenames end with the digits 1, 5, or 9 are reserved for testing, and the remainder are used for training, yielding an approximate 7:3 ratio. Images are used at their original resolution without cropping.

b) CARPK: CARPK is a UAV-collected dataset focused on vehicle detection in parking lot environments, consisting of 1448 images at 1280×720 resolution with 89 777 annotated cars. Images are captured from a consistent top-down viewpoint at low altitude, resulting in a different scale distribution and background context compared to UCAS-AOD. All images are processed at their original resolution without additional cropping, and training and evaluation follow the official dataset split.

B. Implementation Details

RT-DETR is adopted as the base detector with ResNet-34 as the backbone network, setting the input image resolution to 640×640. The model is optimized using AdamW with an initial learning rate of 2×10^{-4}, following the RT-DATR [11] configuration for data augmentation and learning rate scheduling. Training is conducted on two NVIDIA A6000 GPUs with a total batch size of 8, comprising 4 labeled source-domain images and 4 unlabeled target-domain images per iteration. The model is trained for 96 epochs to ensure convergence of the KRD relational alignment objective. For the loss weights in the training objective, the RT-DATR configuration is followed for the shared terms ($\lambda_1 = 1.5$, $\lambda_2 = \lambda_3 = \lambda_4 = 1.0$), and the KRD weight is set to $\lambda_5 = 1.0$. Before domain-adaptive training, the model is pretrained on the source-domain dataset to establish a reliable initialization.

C. Evaluation Metrics

The mean average precision (mAP) at an Intersection-over-Union (IoU) threshold of 0.5 (mAP_{50}) is adopted as the primary evaluation metric, which is standard in DAOD for remote sensing. For each object category, the average precision (AP) is evaluated by summarizing precision and recall across varying confidence thresholds based on the predicted bounding boxes and ground-truth annotations. The mAP_{50} is then computed by averaging the per-category AP values across all evaluated categories. Per-category AP values are additionally reported to enable detailed performance analysis.

D. Comparison With State-of-the-Art Methods

To evaluate the effectiveness of FS^2A, comparisons are made against a broad range of DAOD approaches, including CNN-based methods (DAF, SWF, EPM, HT, FADA, FIE-Net) and Transformer-based methods (SFA, AQT, MRT, RST, DATR, VLDadaptor, RT-DATR). Since all compared methods are evaluated under their originally published backbone configurations, differences in model capacity across methods are inherent to the comparison; the source only and Oracle bounds reported for three base detectors provide a shared reference for contextualizing the absolute performance range. For methods without published results on the evaluated benchmarks, results reported in RST [37] and FIE-Net [14] are adopted as References for remote sensing DAOD evaluation. Source only and Oracle results are additionally reported for three base detectors as performance bounds: Source only denotes training on labeled source data only without adaptation, and Oracle denotes full supervision on target-domain data. These two settings establish the lower and upper bounds for evaluating the benefit of domain adaptation.

1) Results on xView → DOTA: Table I presents results on the cross-satellite benchmark xView → DOTA, where domain shift arises from differences in sensor spectral response, spatial resolution, and annotation granularity. FS^2A achieves 67.6% mAP_{50}, the best overall result and 2.7% above RT-DATR (64.9%), the previous state of the art. At the category level, FS^2A ranks first on ship (67.5%) and storage tank (61.6%), and second on plane (74.1%), where RST (74.9%) holds the lead. The most notable gains appear on ship and storage tank, two small-scale, texture-sensitive categories where cross-domain appearance variation tends to be highly pronounced. Across both CNN-based methods (DAF, SWF, EPM, HT, FADA, FIE-Net) and Transformer-based methods (SFA, AQT, MRT, DATR, VLDadaptor), FS^2A achieves consistent improvements in overall mAP, with the margin over the strongest Transformer-based competitor, DATR (63.1%) reaching 4.5%.

2) Results on UCAS-AOD ↔ CARPK: Table II reports results on the cross-platform vehicle detection benchmarks, where domain shift arises from differences in imaging altitude, object scale, and viewpoint between satellite and UAV sensors. On UCAS-AOD → CARPK, FS^2A achieves 77.6%, ranking second overall behind FIE-Net (79.4%). In the reverse direction CARPK → UCAS-AOD, FS^2A achieves 82.0%, the best result among all compared methods. It outperforms RT-DATR (80.6%) by 1.4% and surpasses FIE-Net (78.9%) and RST

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TABLE I

EXPERIMENTAL RESULTS (%) OF THE SCENARIO BETWEEN DIFFERENT OPTICAL SATELLITES: XVIEW → DOTA

Method	Type	Backbone	Plane	Ship	Storage Tank	mAP_{50}
Source Only Deformable-DETR(ICLR'21) [49]	Transformer	resnet-50	64.8	55.5	32.6	51.0
Oracle Deformable-DETR(ICLR'21) [49]			93.5	74.1	76.2	81.3
Source Only DINO(ICLR'23) [50]	Transformer	resnet-50	59.9	38.8	53.5	50.7
Oracle DINO(ICLR'23) [50]			94.0	78.5	74.4	82.3
Source Only RT-DETR(CVPR'24) [12]	Transformer	resnet-34	68.7	63.3	36.8	56.3
Oracle RT-DETR(CVPR'24) [12]			94.3	74.6	78.4	82.5
DAF(CVPR'18) [6]	CNN	resnet-50	57.3	34.3	36.4	42.7
SWF(CVPR'19) [7]	CNN	resnet-50	48.1	33.9	34.3	38.7
EPM(ECCV'20) [15]	CNN	resnet-101	60.5	59.8	43.9	54.8
SFA(ACM MM'21) [9]	Transformer	resnet-50	65.2	57.1	38.9	53.7
AQT(IJCAI'22) [10]	Transformer	resnet-50	68.2	59.9	46.8	58.3
MRT(ICCV'23) [32]	Transformer	resnet-50	66.3	38.2	32.0	45.5
HT(CVPR'23) [51]	CNN	vgg-16	62.8	55.2	48.8	55.6
FADA(TGRS'22) [8]	CNN	resnet-50	68.0	43.4	46.2	52.5
Remote Sensing Teacher(TGRS'24) [37]	Transformer	resnet-50	**74.9**	58.2	51.5	61.5
DATR(TIP'25) [52]	Transformer	resnet-50	67.8	64.0	<u>57.5</u>	63.1
FIE-Net(TGRS'25) [14]	CNN	vgg-16	66.8	52.6	45.9	55.1
VLDadaptor(TMM'24) [41]	CNN	resnet-50	60.5	57.7	53.9	57.4
RT-DATR [11]	Transformer	resnet-34	72.8	<u>65.8</u>	56.1	<u>64.9</u>
Ours	Transformer	resnet-34	<u>74.1</u>	**67.5**	**61.6**	**67.6**

- *Highest result is shown in **bold**, while second best is <u>underlined</u>.*

TABLE II

EXPERIMENTAL RESULTS (%) ON SATELLITE-TO-UAV (UCAS-AOD → CARPK) AND UAV-TO-SATELLITE (CARPK → UCAS-AOD) BENCHMARKS

Method	Type	Backbone	UCAS-AOD → CARPK	CARPK → UCAS-AOD
Source Only Deformable-DETR(ICLR'21) [49]	Transformer	resnet-50	61.2	58.1
Oracle Deformable-DETR(ICLR'21) [49]			80.9	79.1
Source Only DINO(ICLR'23) [50]	Transformer	resnet-50	52.8	58.4
Oracle DINO(ICLR'23) [50]			76.4	80.9
Source Only RT-DETR(CVPR'24) [12]	Transformer	resnet-34	70.3	67.7
Oracle RT-DETR(CVPR'24) [12]			81.0	92.4
DAF(CVPR'18) [6]	CNN	resnet-50	68.4	53.7
SWF(CVPR'19) [7]	CNN	resnet-50	58.3	57.1
EPM(ECCV'20) [15]	CNN	resnet-101	71.2	65.0
SFA(ACM MM'21) [9]	Transformer	resnet-50	49.8	66.1
AQT(IJCAI'22) [10]	Transformer	resnet-50	71.0	70.8
HT(CVPR'23) [51]	CNN	vgg-16	72.5	67.8
Remote Sensing Teacher(TGRS'24) [37]	Transformer	resnet-50	76.2	75.6
DATR(TIP'25) [52]	Transformer	resnet-50	76.4	73.1
FIE-Net(TGRS'25) [14]	CNN	vgg-16	**79.4**	78.9
VLDadaptor(TMM'24) [41]	CNN	resnet-50	74.4	63.9
RT-DATR [11]	Transformer	resnet-34	75.3	<u>80.6</u>
Ours	Transformer	resnet-34	<u>77.6</u>	**82.0**

- *Highest result is shown in **bold**, while second best is <u>underlined</u>.*

(75.6%) by substantial margins, demonstrating robust cross-platform generalization in both transfer directions.

E. Ablation Studies

To validate the contribution of each proposed module, systematic ablation experiments are conducted on the xView → DOTA benchmark. Unless otherwise stated, all ablation variants are derived from the full FS^2A model by modifying or removing a single component.

1) Component Ablation: Table III shows that both SCFM and KRD contribute independently, and their combination consistently outperforms either module deployed in isolation.

TABLE III

COMPONENT ABLATION ON XVIEW → DOTA (%)

SCFM	KRD	Plane	Ship	Storage Tank	mAP_{50}
		72.8	65.8	56.1	64.9
✓		70.7	66.7	60.3	65.9
	✓	73.7	67.4	56.0	65.7
✓	✓	74.1	67.5	61.6	67.6

SCFM alone improves mAP_{50} by 1.0% over the baseline; KRD alone contributes 0.8%. The full model achieves a 2.7% gain, which exceeds the sum of the two individual improvements

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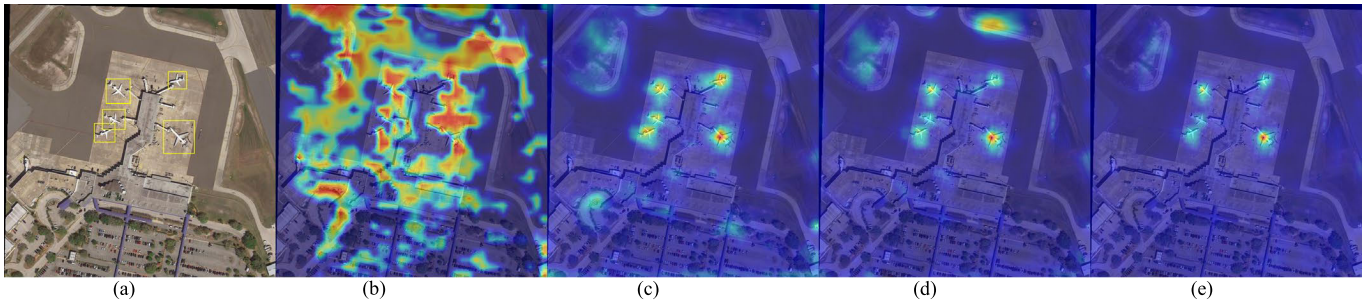


Fig. 5. Grad-CAM activation maps on a target-domain sample (xView→DOTA). (a) GT. (b) Baseline: diffuse, background-dominated activations. (c) +SCFM: reduced background noise. (d) +KRD: sharpened foreground responses. (e) FS^{2}A: most concentrated foreground activation, confirming the complementary effect of both modules.

TABLE IV
ABLATION ON SCFM LOW-FREQUENCY RATIO ON XVIEW → DOTA (%)

Setting	Plane	Ship	Storage Tank	mAP$_{50}$
$r = 0.10$	72.2	66.6	58.6	65.7
$r = 0.15$ (default)	74.1	67.5	61.6	67.6
$r = 0.20$	72.5	65.4	59.3	65.7

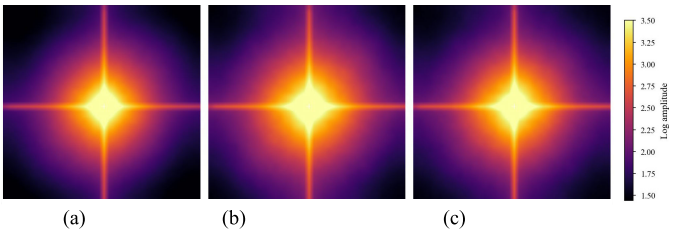


Fig. 6. 2-D log-amplitude spectrum of backbone features (xView→DOTA). (a) Source. (b) Baseline target: visible low-frequency amplitude gap. (c) FS^{2}A target: gap reduced by 42.3%.

(1.8%), indicating that the two modules address different aspects of the domain gap and reinforce each other.

The quantitative results are complemented with Grad-CAM visualizations (Fig. 5) to verify that the mAP gains correspond to accurate foreground localization on target-domain samples. The baseline, Fig. 5(b) produces diffuse activations across background regions, indicating that the model struggles to localize foreground objects under domain shift. SCFM alone, Fig. 5(c) reduces background noise but foreground attention remains incomplete; KRD alone, Fig. 5(d) sharpens foreground responses but some scattered activations persist. The full FS^{2}A model, Fig. 5(e) achieves the most concentrated foreground activation, qualitatively consistent with its best mAP$_{50}$ in Table III.

2) Effectiveness of SCFM: Fig. 6 further illustrates how SCFM narrows the low-frequency amplitude gap between the source and target domains. To study the sensitivity of SCFM to its key design parameter, the low-frequency ratio r that controls the bandwidth of the Gaussian mask is varied, with results in Table IV. A narrowband ($r = 0.10$) misses part of the sensor-dependent spectral variation, limiting mAP$_{50}$ to 65.7%. The default $r = 0.15$ covers the amplitude gap identified in the power spectrum without reaching into midfrequency components, giving the best result of 67.6%. An expanded band ($r = 0.20$) extends modulation into midfrequency content and drops back to 65.7%, confirming that the bandwidth needs to be set carefully.

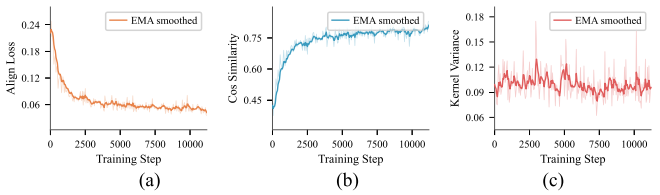


Fig. 7. KRD training dynamics on xView → DOTA. (a) Alignment loss decreases from ~0.24 to ~0.06. (b) Cosine similarity between the student and teacher rises from ~0.45 to ~0.80. (c) Kernel variance stays near ~0.10 throughout, showing no feature collapse.

TABLE V
ABLATION ON KRD HYPERPARAMETERS ON XVIEW → DOTA (%)

Setting	Plane	Ship	Storage Tank	mAP$_{50}$
Dynamic weighting exponent η				
$\eta = 1.0$	73.5	67.3	57.8	66.2
$\eta = 1.5$ (default)	74.1	67.5	61.6	67.6
$\eta = 2.0$	73.0	66.4	59.1	66.2
$\eta = 3.0$	69.9	65.6	60.6	65.4
Polynomial kernel degree d				
$d = 1$ (linear)	72.8	66.4	57.9	65.7
$d = 3$ (default)	74.1	67.5	61.6	67.6
$d = 5$	73.9	63.5	61.6	66.3

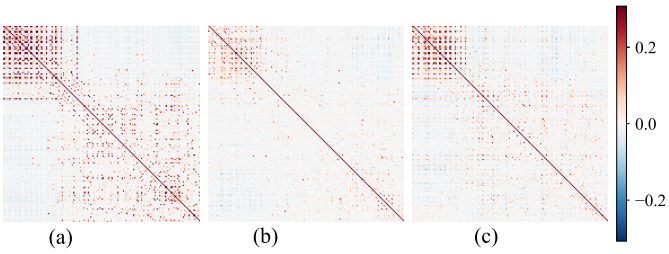


Fig. 8. Polynomial kernel similarity matrices on xView → DOTA. Rows and columns denote spatial tokens flattened in raster-scan order. (a) Teacher (G_{teacher}): clear local block structure reflecting coherent spatial-token relations. (b) Baseline student (centered kernel alignment, CKA = 0.542): weak off-diagonal structure, indicating limited relational alignment with the teacher. (c) FS^{2}A student (CKA = 0.646, ↑ 0.104): teacher-like local relational patterns are partially recovered.

3) Effectiveness of KRD: To verify that KRD progressively transfers relational structure from the teacher to the student,

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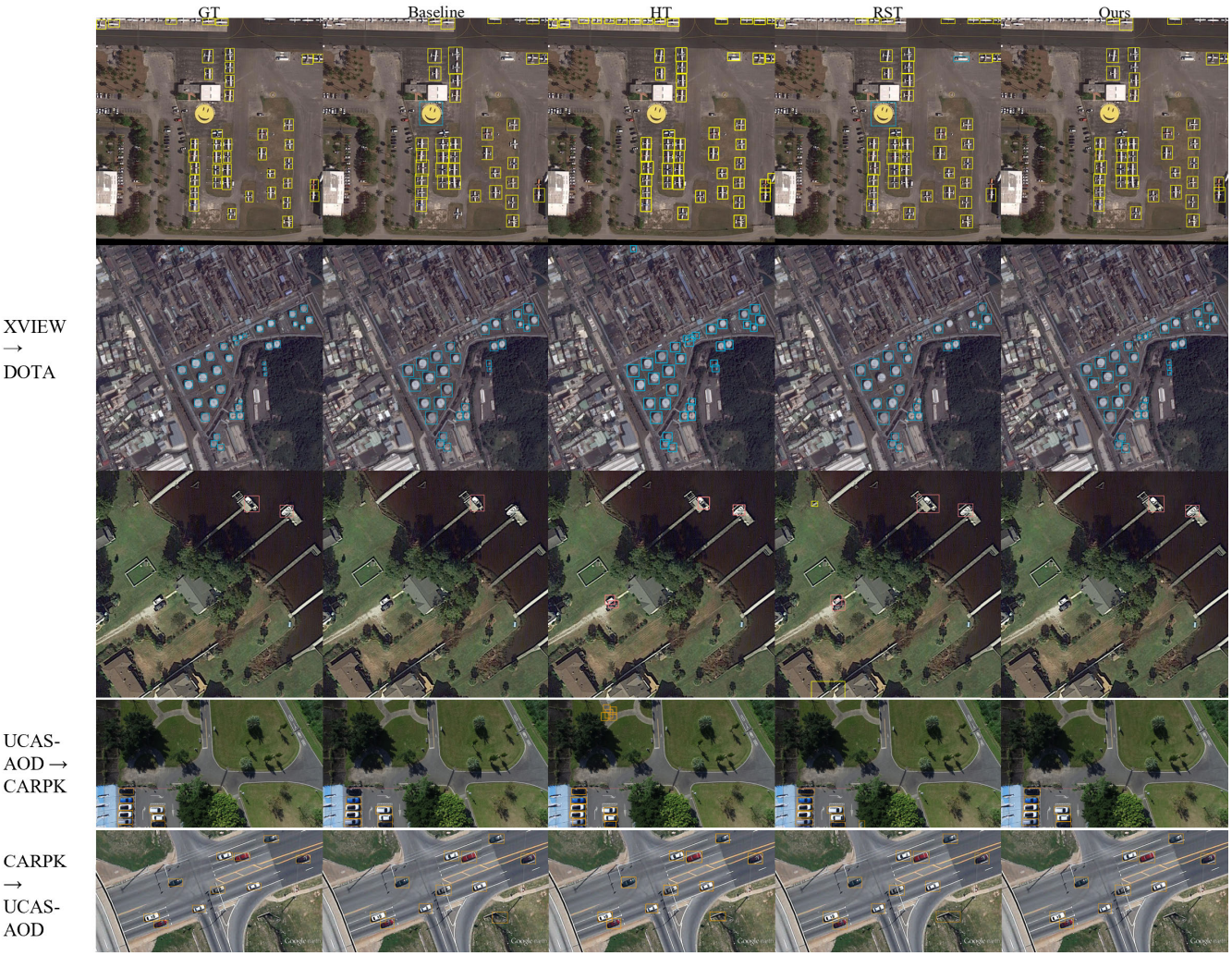


Fig. 9. Qualitative detection results across three benchmarks. Each column shows, from left to right, GT, Baseline, HT, RST, and Ours. Rows 1–3: xView → DOTA, covering planes (yellow), storage tanks (blue), and ships (red). Row 4: UCAS-AOD → CARPK vehicles (orange). Row 5: CARPK → UCAS-AOD vehicles (orange).

three training metrics on xView → DOTA are monitored, as shown in Fig. 7. The alignment loss drops from ∼0.24 to ∼0.06 over 10k steps, showing that the student steadily reduces its gap with the frozen teacher. The cosine similarity rises from ∼0.45 to ∼0.80, confirming that the two representations become highly aligned as training progresses. At the same time, the kernel variance stays near ∼0.10 throughout, showing that the alignment process does not cause feature collapse.

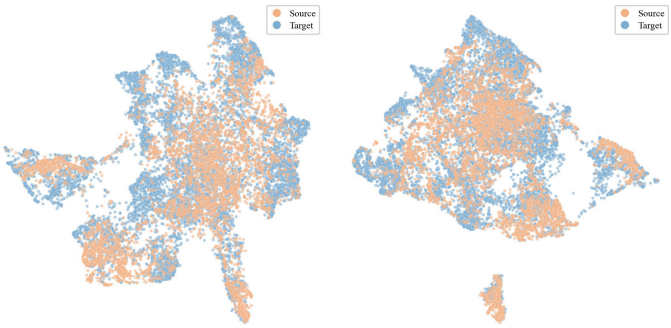
The structural recovery is also visible in the polynomial kernel similarity matrices (Fig. 8). The teacher matrix Fig. 7(a) exhibits clear local relational structure, where locally coherent token groups form stronger pairwise relations. This observation is consistent with prior findings that self-supervised Vision Transformers can group visually similar regions in the embedding space [18]. After KRD training, the FS^{2}A student Fig. 7(c) shows a teacher-like local relational pattern with higher CKA (0.646, ΔCKA = + 0.104), indicating that the kernel transfer objective helps reduce the structural gap between the student and the teacher.

To examine the robustness of KRD to its design choices, results under different settings of η and d are reported in Table V. For η: a linear schedule (η = 1.0) relaxes the constraint too slowly, overconstraining the detection head in later training; elevated values (η = 2.0, 3.0) release the constraint too early, shortening the effective alignment window; the default η = 1.5 strikes the best balance, achieving 67.6% mAP$_{50}$ by reducing late-training interference while maintaining sufficient gradient during midalignment to drive structural transfer. For d: the linear kernel (d = 1) preserves raw pairwise similarities but lacks foreground-selective amplification, so background-dominated token pairs still contribute substantially, and performance drops to 65.7%. The cubic kernel (d = 3) is the lowest odd-degree nonlinear kernel: it preserves similarity polarity while sufficiently amplifying high-similarity foreground pairs, yielding the best performance of 67.6%. In contrast, the quintic kernel (d = 5) makes the relational constraint overly sharp by overamplifying high-similarity pairs, reducing robustness and degrading performance to 66.3%.

F. Visualization and Analysis

1) Detection Results: Fig. 9 presents qualitative comparisons across three remote sensing benchmarks. On the xView → DOTA benchmark (Rows 1–3), the baseline misses

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(a) (b)

Fig. 10. UMAP visualization of feature distributions for the source domain (orange) and the target domain (blue). (a) Baseline model. (b) Ours.

multiple targets and produces false positives on background patterns; HT overpredicts, generating numerous spurious boxes; RST produces false positives on background regions and still suffers from missed detections. Dense storage tanks (Row 2) challenge all three comparison methods, which uniformly exhibit missed detections. In the ship scenario (Row 3), background clutter is particularly damaging: the baseline misses ships on water, HT misclassifies empty parking spaces as ships, and RST misdetects roadside vehicles as ships. On the vehicle detection benchmarks (Rows 4 and 5, UCAS-AOD → CARPK and CARPK → UCAS-AOD), the baseline misses small-scale targets, while HT and RST improve recall at the cost of increased false positives from road markings and vegetation. Across all three benchmarks, FS^{2}A achieves the best balance between foreground recall and background suppression, with predictions most closely matching the ground truth (GT).

2) UMAP Feature Visualization: To visually assess feature alignment across domains, UMAP is applied to reduce the high-dimensional features to a 2-D embedding (Fig. 10). The baseline model, Fig. 10(a) produces clearly separated source (orange) and target (blue) distributions: the periphery of the embedding and several local clusters are dominated by samples from a single domain, indicating that the two domains fail to mix and that the baseline struggles to bridge the cross-domain distribution gap. In contrast, FS^{2}A, Fig. 10(b) substantially closes the gap between the two distributions. Feature points from both domains are densely interleaved across the embedding space, with domain separation visibly reduced, demonstrating that FS^{2}A suppresses domain discrepancy and encourages robust domain-invariant features.

V. Conclusion

Unsupervised DAOD in remote sensing suffers from two alignment limitations: cross-domain appearance variation, where differences in imaging conditions produce divergent visual appearances across domains, and foreground-background imbalance, where the extreme sparsity of targets allows background statistics to dominate alignment. FS^{2}A, a dual-level selective alignment framework built on RT-DATR, addresses both limitations through two complementary modules. At the frequency level, SCFM decomposes multiscale features via FFT and restricts adversarial modulation to the low-frequency amplitude conditioned on scale level and categorical composition, while preserving the domain-invariant phase spectrum. At the spatial level, KRD distills pairwise relational structure from a frozen satellite-pretrained VFM in polynomial kernel space, where polynomial kernel functions preferentially concentrate alignment on foreground feature pairs over weakly correlated background pairs.

Experiments on two cross-domain remote sensing benchmarks validate the effectiveness of FS^{2}A. On the cross-satellite benchmark, xView→DOTA, FS^{2}A achieves 67.6% mAP$_{50}$, surpassing the previous best RT-DATR by 2.7 points. On the satellite-to-UAV benchmarks, FS^{2}A ranks second on UCAS-AOD→CARPK (77.6%) and achieves the best result on CARPK→UCAS-AOD (82.0%), outperforming RT-DATR by 1.4 points. Ablation studies confirm that SCFM and KRD address complementary aspects of the domain gap, and their combination consistently outperforms either module alone.

Extending the framework to source-free DAOD remains an important direction for future work. This setting is particularly relevant to remote sensing applications where source imagery may be unavailable during adaptation due to storage, privacy, or data-sharing constraints.

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